Sarthak Pati

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Education

Technical University of Munich — Munich, Germany
Ph.D., Computer Science, Summa cum laude — 2025

Technical University of Munich — Munich, Germany
M.S., Biomedical Computing — 2014

Manipal Academic of Higher Education — Manipal, India
B.E., Biomedical Engineering — 2010

Summary

Founder and AI safety researcher with **15+** years building trustworthy, end-to-end AI systems. Founder of **VerySafe.ai**, developing **SafeCompute**, a policy-aware compute platform that attaches cryptographic proof to every AI model run, enabling regulated deployment of frontier and open-source LLMs. Track record of **USD 9M+** in NIH-funded research and high-impact publications in *Nature*, *Nature Communications*, and *IEEE TMI*. Passionate about bridging frontier AI capability with the safety, attestation, and governance infrastructure required to deploy it responsibly.

Technical Skills

Python/Rust/C++	Confidential Computing	Foundational Models / LLMs
PyTorch/TensorFlow	Cryptographic Attestation	LLM Training & Routing
Docker/Kubernetes	GitHub Actions / MLOps	Policy & Governance

Latest Professional Experience

VerySafe AI — Miami, FL
Founder — 01/2026 - Present

- Building **SafeCompute**, a policy-aware compute platform that attaches cryptographic proof (RATS attestation, SLSA provenance, HSM-signed audit chains) to every model run, enabling regulated LLM deployment.
- Built a hybrid LLM router selecting across frontier and open-source models on policy constraints (data residency, sensitivity tier, required certifications), with every decision cryptographically logged for auditor-grade lineage.

Vaiyu Solutions — Miami, FL
Owner and CEO — 10/2023 - Present

- Providing specialized consulting related to AI operationalization for **4+** clients of various sizes.
- Managing an interdisciplinary team of **5** engineers and scientists to deliver end-to-end AI model design, training, deployment, and monitoring solutions across pharma, healthcare, finance, and energy domains.
- Driving AI optimization during pre-training phases to reduce training costs by up to **50%** while improving accuracy.
- Implementing scalable AI solutions through API-driven architectures by leveraging open standards (REST, MCP, A2A), and applying modern MLOps practices to translate cutting-edge AI advancements into impactful business outcomes.

Indiana University — Indianapolis, IN (Remote)
Software Architect — 09/2023 - 09/2025

- Led the creation of "data as IP" and "model as IP" strategies to push R&D efforts into privacy and security applications, culminating in a **USD 3.5 million** grant from NIH/NCI for its research.
- Led the design and development of **8+** projects across inter-disciplinary domains such as healthcare AI, privacy, security, federated learning, optimization, and benchmarking.
- Integrated standardized model training (GaNDLF) to reduce prototyping time by **30%** and optimization routines for model inference to reduce resource requirements between **10-50%** and reduced overall inference latency up to **70%**.
- Created and pushed an organization-level strategy to adopt latest research faster by including native support of latest open-source libraries (such as transformers, lightning, mlflow) in HPC compute stack, reducing the amount of custom research environments and containers needed by up to **20%**.

University of Pennsylvania — Philadelphia, PA
Application Architect — 02/2023 - 08/2023

- Led the R&D of a **USD 1.2 million** grant from NIH to deploy federated learning in real-world healthcare infrastructure.
- Streamlined **10+** legacy projects to improve maintainability and reducing the technical debt over time.

University of Pennsylvania — Philadelphia, PA
Senior Application Developer — 12/2014 - 01/2023

- Led the R&D of a **USD 5 million** grant from NIH/NCI to opertionalize AI for effective use by clinicians.
- Led the software development efforts for a team of **5** developers and **25** researchers

Honors and Awards

- Dean's list for doctoral dissertation (top **10%**) 2020-2024.
- Nature Communications Engineering **Editor's choice** 2023 for "Generally Nuanced Deep Learning Framework" [ref].
- Nature Communications Top **25** Health Sciences Articles 2022 for "Federated Learning in Healthcare" [ref].
- Plenary presentation (top **8**) at Pendergrass Symposium 2023 for "Comprehensive Federated Ecosystem (COFE)".
- Best poster award (top **5%**) at NIH Annual Scientific Meeting of the ITCR funding program 2020 and 2022.
- Oral Presentation (top **5%**) at Pendergrass Symposium 2022 for "AI-based Volumetric Breast Density Estimation with Digital Breast Tomosynthesis".
- Oral Presentation (top **5%**) at Pendergrass Symposium 2021 for "Federated Tumor Segmentation".
- Magna cum laude (top **10%**) at Pendergrass Symposium 2021 for "Generally Nuanced Deep Learning Framework".
- **2nd place** in the Automatic Non-Rigid Histological Image Registration Challenge 2019 [ref].
- **1st place** in the Brain Tumor Segmentation Challenge at MICCAI 2015.

Media Mentions of Research

- Wall Street Journal
- Eureka Alert
- News Medical
- IU News
- Science Daily

Invited Talks (Full List)

- Tutorials (half and full day) on Federated Learning at multiple top-tier AI and clinical conferences:
  - Medical Image Computing and Computer Assisted Intervention (MICCAI)
  - Association for the Advancement of Artificial Intelligence (AAAI)
  - Radiological Society of North America (RSNA)
  - Society for Optics and Photonics (SPIE) Medical Imaging
- Presentation at MRI Together Conference.
- Presentation at University of Edinburgh.
- Multiple presentations at Georgetown University.
- Presentation and demonstration of applications of radiomics and federated learning applications healthcare at the Radiological Society of North America (RSNA) Annual Meet (2016-2021).
- Half Day Tutorial on Cancer Imaging at the IEEE International Symposium on Biomedical Imaging (ISBI) (2018).
- Presentation and demonstration of radiomics integration in healthcare (CaPTk) at the International Society for Optics and Photonics (SPIE) Medical Imaging Conferences (2017-2019).

Service

- Serving as Vice Chair for Algorithmic Development at the Medical Working Group of MLCommons since 2023, a non-profit aiming to improve machine learning for the community.
- Academic reviewer for **10+** journals (IEEE TMI, Nature Communications) and conferences (NeurIPS, MICCAI, SPIE).
- Teaching concepts related scientific programming, production-level machine learning in academic conferences.
- Maintainer of **10+** open-source projects and **40+** conda recipes.
- Active open-source contributor to **50+** projects.

Selected Publications (Full List)

1. **S. Pati**, S. Wagner, et al. An Unsupervised Brain Extraction Quality Control Approach for Efficient Neuro-Oncology Studies. (***Journal of Imaging Informatics in Medicine***) 2025.
2. S. Thakur, **S. Pati**, et al. Optimization of Deep Learning Models for inference in low resource environments. (***IEEE Computers in Biology and Medicine***) 2025.
3. **S. Pati**, R. Turrisi, et al. Adapting to evolving MRI data: A transfer learning approach for Alzheimer's disease prediction. (***NeuroImage***) 2025.
4. **S. Pati**, U. Baid, et al. Pan-Cancer Tumor Infiltrating Lymphocyte Detection based on Federated Learning. (***IEEE Conference on Big Data***) 2024.
5. **S. Pati**, et al. Privacy preservation for federated learning in health care. (***Cell Patterns***) 2024.